Prime-Gap Ordering, RH-Stability, and the Zeta-Multiplicity Phase Operator: A Resource-Bounded Numerical Protocol and Defensive Publication

Citizen Gardens
The foundation of Multiplicity

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Abstract

We present a mathematically explicit, computationally executable framework for testing *RH-stability*—the claim that dynamical stability of a prime-encoded quantum system is uniquely associated with the Riemann Hypothesis. The central object is a *Zeta-Multiplicity Phase Operator* acting on a finite prime-chain Hilbert space, driven by a time-dependent Hamiltonian whose modulation is built from the non-trivial zeros of the Riemann zeta function. We articulate two falsifiable experiments: (A) sensitivity to prime-gap ordering at fixed drive, and (B) sensitivity to an exponential envelope parameter β associated with off-line perturbations of the zeros. We provide explicit Hamiltonians, observables (circular phase variance, inverse participation ratio, fidelity), ensemble-based statistical tests (Welch's t-test, Cohen's d), and a resource-bounded NumPy/SciPy implementation strategy that treats computational finitude as part of the ontological substrate. This document is intended both as a comprehensive technical description and as defensive prior art for future developments.

1 Executive Summary

1.1 Core idea

We consider a quantum system whose configuration space is indexed by the first N prime numbers. The Hilbert space is

$$\mathcal{H}_N = \mathrm{span}\{|p_i\rangle : p_i \text{ is the } i\text{-th prime},\ i = 1, \ldots, N\}.$$

On this space we define a time-dependent Hamiltonian

$$H(t;\beta) = H_{\text{prime}} + \lambda(t;\beta)\, H_{\text{gap}},$$

where

- H_{prime} encodes a prime-weighted diagonal spectrum,
- H_{gap} encodes hopping along the chain weighted by normalized prime gaps,
- $\lambda(t;\beta)$ is a scalar modulation constructed from the imaginary parts of the first K non-trivial zeros of the Riemann zeta function.

The parameter β enters in an exponential envelope $e^{\beta t}$ multiplying contributions from the zeta zeros. Conceptually, $\beta = 0$ corresponds to all zeros lying on the critical line, whereas $\beta \neq 0$ mimics off-line deformations.

The central claim we make testable is:

RH-stability hypothesis: For $\beta = 0$, certain dynamical observables (e.g. circular phase variance) remain bounded or grow at most sub-exponentially, whereas for $\beta \neq 0$ these observables exhibit exponential divergence or decay.

In addition, we propose that the *ordering* of prime gaps along the chain generates distinct dynamical signatures relative to a null model where the same gaps are randomly permuted.

1.2 Two experiments

We isolate two numerically accessible, falsifiable experiments:

Experiment A: Prime-gap ordering sensitivity. Fix β (e.g. $\beta = 0$). Define a "true" model with normalized gaps $\tilde{g}_i$ in their natural order and a null model where $\tilde{g}_i$ are randomly permuted along the prime index while keeping their marginal distribution fixed. Evolve ensembles of trajectories under the same zeta-driven modulation and compare time-averaged observables for true vs. null ensembles using Welch's t-test and Cohen's d. If the RH-stability picture is correct, the ordered-gap ensemble should exhibit systematically lower phase variance and higher coherence than the shuffled ensemble.

Experiment B: Exponential-envelope RH sweep. Fix (N, K) and other parameters, and vary $\beta \in \{-\beta_0, 0, +\beta_0\}$ with β_0 small (e.g. 0.05). For each β run an ensemble of trajectories, fit linear slopes of $\log V(t)$ and $-\log F(t)$ at late times, and compare slope distributions via Welch's t-test. Under RH-stability, the slopes for $\beta = 0$ should concentrate near zero, while slopes for $\beta \neq 0$ should be significantly non-zero.

1.3 Resource-bounded protocol

We adopt a resource-bounded view: numerical integration is performed with standard SciPy solvers (e.g. DOP853), and only those experiments that complete reliably on realistic local hardware are considered part of the test. In particular:

- We work with small prime truncations (e.g. $N = 15$–40) and relatively few zeros (e.g. $K = 5$–10).
- We use moderate time windows (e.g. $T = 20$–100) and time steps (Δt) chosen to balance resolution with runtime.
- Ensemble sizes (number of realizations per parameter) are kept to a few tens in the baseline tests.

This protocol explicitly acknowledges finite computational resources as part of the "constitutional" structure of the experiment.

1.4 Defensive publication purpose

This document:

- Specifies the Hamiltonians, observables, and statistical comparisons in sufficient detail for independent replication.
- Records the idea that RH-stability can be probed numerically via a finite-dimensional prime-chain Hamiltonian with zeta-zero modulation and exponential envelope.
- Emphasizes the role of prime-gap ordering and ensembles as key discriminants between lawful (critical-line-like) and unstable (off-line-like) dynamics.

This serves as prior art for future theoretical refinements, implementation in alternative frameworks (e.g. QuTiP, tensor networks), and hardware realizations.

2 Comprehensive Mathematical Overview

2.1 Prime-chain Hilbert space and Hamiltonian

Let $p_1 = 2, p_2 = 3, p_3 = 5, \ldots, p_N$ denote the first N prime numbers. Define basis states $|p_i\rangle$ orthonormal in $\mathcal{H}_N$, and a pure state

$$|\psi(t)\rangle = \sum_{i=1}^{N} c_i(t)\,|p_i\rangle\,, \qquad \sum_{i=1}^{N} |c_i(t)|^2 = 1.$$

2.1.1 Prime-diagonal term

We define a diagonal Hamiltonian

$$H_{\text{prime}} = \alpha \sum_{i=1}^{N} \frac{1}{p_i}\,|p_i\rangle\,\langle p_i|\,, \tag{1}$$

where α is a real scaling parameter. This term encodes a simple prime-weighted spectrum.

2.1.2 Gap-hopping term

Let $g_i = p_{i+1} - p_i$ denote adjacent prime gaps for $i = 1, \ldots, N-1$. We define normalized gaps $\tilde{g}_i$ via either:

$$\tilde{g}_i = \frac{g_i}{g_{\max}}, \qquad g_{\max} = \max_j g_j, \tag{2}$$

or

$$\tilde{g}_i = \frac{\log(1+g_i)}{\log(1+g_{\max})}, \tag{3}$$

depending on numerical convenience. The gap-induced hopping Hamiltonian is

$$H_{\text{gap}} = \eta \sum_{i=1}^{N-1} \tilde{g}_i\,(|p_i\rangle\,\langle p_{i+1}| + |p_{i+1}\rangle\,\langle p_i|)\,, \tag{4}$$

with coupling strength η. This couples neighboring primes with amplitudes proportional to the normalized gaps.

2.2 Zeta-zero modulation and exponential envelope

Let $\rho_n = \frac{1}{2} + it_n$ denote the first K non-trivial zeros of the Riemann zeta function (ordered by increasing $t_n > 0$). We use the imaginary parts t_n as frequency-like parameters.

2.2.1 Envelope and drive

We define a scalar modulation

$$\lambda(t;\beta) = 1 + \sum_{n=1}^{K} \gamma_n e^{\beta t} \cos(t_n t + \phi_n), \tag{5}$$

where:

- $\gamma_n = \gamma_0/\sqrt{n}$ is a decaying amplitude profile.
- ϕ_n are random phases drawn independently from $\text{Unif}[0, 2\pi)$ for each realization.
- $\beta \in \mathbb{R}$ is a control parameter: $\beta = 0$ is taken as the "critical-line" case, and $\beta \neq 0$ as off-line deformations.

In numerical implementations one may also impose a lower bound $\lambda_{\min} > 0$ via $\lambda(t) \leftarrow \max(\lambda(t), \lambda_{\min})$ to avoid sign flips or collapse of the hopping term.

2.2.2 Total Hamiltonian and dynamics

The total Hamiltonian is

$$H(t;\beta) = H_{\text{prime}} + \lambda(t;\beta)\, H_{\text{gap}}. \tag{6}$$

We consider unitary evolution governed by

$$i\frac{d}{dt}\,|\psi(t)\rangle = H(t;\beta)\,|\psi(t)\rangle\,, \tag{7}$$

with $\hbar = 1$. For numerical integration we use vectorized form

$$\dot{\psi}(t) = -iH(t;\beta)\psi(t),$$

where $\psi(t) \in \mathbb{C}^N$ is the column vector of amplitudes $c_i(t)$.

2.3 Observables

Given state coefficients $c_i(t)$, we define several observables for each time t.

2.3.1 Circular phase variance

Let $\theta_i(t) = \arg c_i(t) \in (-\pi, \pi]$ denote phases of amplitudes. Define the mean resultant vector:

$$R(t) = \frac{1}{N}\sum_{i=1}^{N} e^{i\theta_i(t)}. \tag{8}$$

The circular mean angle is $\theta_{\text{mean}}(t) = \arg R(t)$. We define wrapped deviations

$$\delta_i(t) = \arg\left(e^{i(\theta_i(t)-\theta_{\text{mean}}(t))}\right) \in (-\pi, \pi], \tag{9}$$

and the circular phase variance

$$V(t) = \frac{1}{N}\sum_{i=1}^{N} \delta_i(t)^2. \tag{10}$$

This is invariant under global phase shifts and robust to phase wrapping.

2.3.2 Inverse participation ratio (IPR)

The IPR quantifies localization in the prime basis:

$$L(t) = \sum_{i=1}^{N} |c_i(t)|^4. \tag{11}$$

A large $L(t)$ indicates localization on a few basis states; a small $L(t)$ indicates delocalization.

2.3.3 Fidelity / Loschmidt echo

Given initial state $|\psi_0\rangle$, commonly chosen as the uniform superposition

$$|\psi_0\rangle = \frac{1}{\sqrt{N}}\sum_{i=1}^{N} |p_i\rangle\,,$$

we define the time-dependent fidelity

$$F(t) = |\,\langle\psi_0|\psi(t)|\psi_0|\psi(t)\rangle\,|^2. \tag{12}$$

This is a Loschmidt echo measuring global recurrence.

2.4 Ensemble definitions

Randomness enters through the phases $\{\phi_n\}$ in the drive. For fixed $(N, K, \alpha, \eta, \gamma_0, \beta)$, we define an ensemble of trajectories by sampling independent $\{\phi_n\}$ and integrating.

For **Experiment A** (gap ordering), we define:

- **True ensemble:** Use the normalized gaps $\tilde{g}_i$ derived from the actual prime sequence in ascending order.
- **Null ensemble:** For each realization, draw a random permutation π of $\{1, \ldots, N-1\}$ and set $\tilde{g}_i^{(\text{null})} = \tilde{g}_{\pi(i)}$, while keeping the same sets of primes $\{p_i\}$ and zero frequencies $\{t_n\}$.

For **Experiment B** (envelope sweep), we fix gap ordering and vary β.

2.5 Time-averaged quantities and slope extraction

2.5.1 Time-averaged observables (Experiment A)

To reduce transient effects we define time-averaged observables over a window $[T/2, T]$:

$$\bar{V} = \frac{2}{T}\int_{T/2}^{T} V(t)\,dt, \qquad \bar{L} = \frac{2}{T}\int_{T/2}^{T} L(t)\,dt, \qquad \bar{F} = \frac{2}{T}\int_{T/2}^{T} F(t)\,dt. \tag{13}$$

In numerical practice these are approximated by discrete averages over the sampled time points in the latter half of the evolution.

2.5.2 Slope fitting for exponential behavior (Experiment B)

To detect exponential growth or decay we fit linear slopes to $\log V(t)$ and $\log F(t)$ at late times. Given discrete times t_k and observable values $Y_k \equiv Y(t_k)$, we select a fitting window $t \in [t_{\text{start}}, T]$ and perform least-squares regression:

$$\log Y_k \approx s_Y t_k + c_Y.$$

The slope s_Y is the estimated exponential growth rate:

$$Y(t) \approx e^{s_Y t + c_Y}. \tag{14}$$

We denote by s_V and s_F the slopes for V and F (or for $-\log F$, depending on convention).

3 Statistical Testing and Falsifiability

3.1 Welch's t-test

For Experiment A, we obtain samples of time-averaged observables:

$$\{\bar{V}_j^{(\text{true})}\}_{j=1}^{n_{\text{true}}}, \quad \{\bar{V}_r^{(\text{null})}\}_{r=1}^{n_{\text{null}}}$$

and similarly for $\bar{L}, \bar{F}$. To compare means we use Welch's two-sample t-test, which does not assume equal variances.

Let X denote any of these observables (e.g. $\bar{V}$). Define sample means $\bar{X}_1, \bar{X}_2$ and sample variances s_1^2, s_2^2 for true and null ensembles with sizes n_1, n_2. The Welch statistic is

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}},$$

with degrees of freedom

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{s_1^4}{n_1^2(n_1-1)} + \frac{s_2^4}{n_2^2(n_2-1)}}.$$

We report the two-sided p-value associated with this statistic.

3.2 Effect size: Cohen's d

To quantify effect size we define Cohen's d:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pooled}}},$$

with pooled standard deviation

$$s_{\text{pooled}} = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}.$$

Moderate to large effects correspond to $|d| \gtrsim 0.8$, though thresholds depend on context.

3.3 Falsifiability statements

3.3.1 Experiment A: Gap-ordering stability

We say that the model exhibits *gap-ordering stability* for a given parameter setting if there exists a choice of $(N, K, \alpha, \eta, \gamma_0)$ such that:

- $\bar{V}^{(\text{true})} < \bar{V}^{(\text{null})}$ with $p < 0.05$ and $|d_V| \gtrsim 0.8$,
- and the signs of $\Delta\bar{L}$ and $\Delta\bar{F}$ are consistent with increased localization and coherence under true ordering (e.g. $\bar{L}^{(\text{true})} \geq \bar{L}^{(\text{null})}$, $\bar{F}^{(\text{true})} \geq \bar{F}^{(\text{null})}$).

The hypothesis that prime-gap ordering matters dynamically is *falsified* if, after scanning reasonable parameter grids, one finds no region where the true vs. null difference is statistically significant and effect sizes remain small across all observables.

3.3.2 Experiment B: RH-stability under envelope

We define RH-stability in this model as the condition that:

- For $\beta = 0$, the ensemble-averaged slopes satisfy $\mathbb{E}[s_V(0)] \approx 0$ and $\mathbb{E}[s_V(0)]$ is statistically distinguishable from $\mathbb{E}[s_V(\beta_0)]$ for $\beta_0 \neq 0$ (e.g. $\beta_0 = 0.05$),
- and similarly for s_F (using either $\log F$ or $-\log F$).

The RH-stability hypothesis in this formulation is *falsified* if the slope distributions for $\beta = 0$ and $\beta \neq 0$ cannot be distinguished at a chosen significance level (e.g. $p < 0.01$) for any reasonable parameter region.

4 NumPy/SciPy Implementation

4.1 Design goals

The implementation is designed to:

- Use only widely available numerical libraries (NumPy, SciPy).
- Avoid specialized quantum packages to remain environment-agnostic.
- Separate model definition, ensemble generation, slope fitting, and hypothesis testing.

4.2 Core code snippet

Below is a consolidated Python code snippet illustrating the main components. For brevity, some comments are minimized; in practice you may want to add docstrings and logging.

```
import numpy as np
from scipy.integrate import solve_ivp
from scipy.stats import ttest_ind

# ---------- basic utilities ----------
def first_n_primes(N):
    primes = []
    x = 2
    while len(primes) < N:
        for p in primes:
            if p * p > x:
                break
            if x % p == 0:
                break
        else:
            primes.append(x)
        x += 1
    return np.array(primes, dtype=int)

def prime_gaps(primes):
    return np.diff(primes)

def normalize_gaps(gaps, mode="log"):
    gaps = np.array(gaps, dtype=float)
    if mode == "max":
        return gaps / gaps.max()
    if mode == "log":
        return np.log1p(gaps) / np.log1p(gaps.max())
    raise ValueError("mode must be 'max' or 'log'")

def first_zero_imag_parts(K):
    t_vals = np.array([
        14.134725, 21.022040, 25.010858, 30.424876, 32.935062,
        37.586178, 40.918719, 43.327073, 48.005151, 49.773832,
        52.970321, 56.446248, 59.347044, 60.831779, 65.112544,
        67.079811, 69.546402, 72.067158, 75.704691, 77.144840
    ])
    return t_vals[:K]

# ---------- Hamiltonian construction ----------
def build_prime_matrix(primes, alpha):
    return np.diag(alpha / primes.astype(float)).astype(complex)

def build_gap_matrix(gnorm, eta):
    N = len(gnorm) + 1
    H = np.zeros((N, N), dtype=complex)
    for i in range(N - 1):
```

```
        c = eta * gnorm[i]
        H[i, i + 1] = c
        H[i + 1, i] = c
    return H

# ---------- time-dependent coefficient ----------
def lambda_coeff(t, zero_freqs, gammas, phases, beta):
    s = 1.0
    for w, g, phi in zip(zero_freqs, gammas, phases):
        s += g * np.exp(beta * t) * np.cos(w * t + phi)
    return s

def rhs(t, psi, H_prime, H_gap, zero_freqs, gammas, phases, beta):
    lam = lambda_coeff(t, zero_freqs, gammas, phases, beta)
    Ht = H_prime + lam * H_gap
    return -1j * (Ht @ psi)

def run_single_beta(primes, gnorm, zero_freqs, gamma_scale, beta,
                    alpha=1.0, eta=0.3, T=20.0, dt=0.1):
    N = len(primes)
    H_prime = build_prime_matrix(primes, alpha)
    H_gap = build_gap_matrix(gnorm, eta)
    psi0 = np.ones(N, dtype=complex) / np.sqrt(N)
    K = len(zero_freqs)
    gammas = gamma_scale / np.sqrt(np.arange(1, K + 1))
    phases = np.random.uniform(0, 2*np.pi, size=K)
    t_eval = np.arange(0, T + dt, dt)
    sol = solve_ivp(
        fun=lambda t, y: rhs(t, y, H_prime, H_gap,
                             zero_freqs, gammas, phases, beta),
        t_span=(0.0, T),
        y0=psi0,
        t_eval=t_eval,
        rtol=1e-7,
        atol=1e-9,
        method='DOP853'
    )
    states = sol.y.T
    t = sol.t
    # circular phase variance
    angles = np.angle(states)
    mean_vec = np.mean(np.exp(1j * angles), axis=1)
    circ_mean_angle = np.angle(mean_vec)
    dev = angles - circ_mean_angle[:, None]
    dev = np.angle(np.exp(1j * dev))
    V = np.mean(dev**2, axis=1)
    # fidelity
    F = np.abs(states @ np.conjugate(psi0))**2
    return {"t": t, "V": V, "F": F, "beta": beta}

def fit_log_slope(t, y, t_start=5.0, t_end=None):
```

```
    mask = (t >= t_start)
    if t_end is not None:
        mask &= (t <= t_end)
    t_masked = t[mask]
    y_masked = y[mask]
    y_masked = np.maximum(y_masked, 1e-12)
    logy = np.log(y_masked)
    A = np.vstack([t_masked, np.ones_like(t_masked)]).T
    slope, const = np.linalg.lstsq(A, logy, rcond=None)[0]
    return slope

def run_ensemble(beta, num_realizations=5, N=15, K=5,
                 alpha=1.0, eta=0.3, gamma_scale=0.1,
                 T=20.0, dt=0.1, t_start_fit=5.0, seed=42):
    np.random.seed(seed)
    primes = first_n_primes(N)
    gaps = prime_gaps(primes)
    gnorm = normalize_gaps(gaps, mode="log")
    zero_freqs = first_zero_imag_parts(K)
    slopes_V, slopes_F = [], []
    for _ in range(num_realizations):
        traj = run_single_beta(primes, gnorm, zero_freqs,
                               gamma_scale, beta,
                               alpha, eta, T, dt)
        sV = fit_log_slope(traj["t"], traj["V"], t_start_fit, T)
        sF = fit_log_slope(traj["t"], traj["F"], t_start_fit, T)
        slopes_V.append(sV)
        slopes_F.append(sF)
    slopes_V = np.array(slopes_V)
    slopes_F = np.array(slopes_F)
    return {
        "beta": beta,
        "slopes_V": slopes_V,
        "slopes_F": slopes_F,
        "mean_V": slopes_V.mean(),
        "std_V": slopes_V.std(ddof=1),
        "mean_F": slopes_F.mean(),
        "std_F": slopes_F.std(ddof=1),
    }

def run_rh_stability_test():
    betas = [-0.05, 0.0, 0.05]
    results = {}
    for beta in betas:
        print(f"Running beta = {beta} ...")
        res = run_ensemble(beta)
        results[beta] = res
        print(f" mean slope V = {res['mean_V']:.4f} ± {res['std_V']:.4f}")
        print(f" mean slope F = {res['mean_F']:.4f} ± {res['std_F']:.4f}")
    # example comparison: beta=0 vs beta=0.05
    slopes_V_null = results[0.0]["slopes_V"]
```

```
    slopes_V_pos  = results[0.05]["slopes_V"]
    t_stat, p_val = ttest_ind(slopes_V_null, slopes_V_pos,
                              equal_var=False)
    print("\nWelch t-test (beta=0 vs beta=0.05): "
          f"t = {t_stat:.3f}, p = {p_val:.3g}")
    return results

if __name__ == "__main__":
    run_rh_stability_test()
```

This script is an ultra-reduced baseline for Experiment B (envelope sweep). A similar pattern can be adopted for Experiment A (gap ordering) by introducing separate functions for true and permuted gap matrices and comparing time-averaged observables instead of slopes.

5 Prior Art and Defensive Claims

5.1 Key claims of originality

The following points summarize what this document asserts as prior art:

1. **Prime-chain RH-stability via exponential envelope.** The idea of encoding RH-stability as bounded vs. exponential behavior in phase variance and fidelity under a Hamiltonian of the form

$$H(t;\beta) = H_{\text{prime}} + \lambda(t;\beta)H_{\text{gap}}, \quad \lambda(t;\beta) = 1 + \sum_n \gamma_n e^{\beta t}\cos(t_n t + \phi_n),$$

   with the t_n drawn from the non-trivial zeta zeros, in a finite prime-chain Hilbert space.

2. **Gap-ordering ensemble test.** The construction of a true vs. null ensemble based on ordered vs. permuted normalized gaps $\tilde{g}_i$, preserving the marginal gap distribution while destroying ordering, and the use of Welch's t-test and Cohen's d on time-averaged observables to quantify the dynamical effect of prime-gap ordering.

3. **Exponential-slope ensemble test for β.** The use of ensemble slope distributions of $\log V(t)$ (and/or $-\log F(t)$) across different β values, compared via Welch's test, as a direct numerical probe linking RH-like conditions ($\beta = 0$) to stability and off-line conditions ($\beta \neq 0$) to exponential divergence.

4. **Resource-bounded ontic protocol.** The explicit treatment of computational finitude (e.g. small N, K, short T, tractable ensemble sizes) as part of the ontological structure of the test, and the associated insistence that only locally executed, user-verifiable runs provide admissible numerical evidence.

5.2 Intended uses

This publication is designed to be:

- A reference for future theoretical extensions (e.g. regularization of Euler products, inclusion of multiplicative projectors, tensor-network formulations).
- A baseline for experimental proposals mapping H_{prime} and H_{gap} to controllable frequencies and couplings in physical platforms (e.g. superconducting qubits).
- A template for ensemble-based numerical investigations of RH-related conjectures using finite-dimensional dynamical systems.

6 Conclusion

We have specified a complete, finite-dimensional dynamical model that ties prime gaps, zeta zeros, and RH-stability into a single, executable framework. Two falsifiable experiments—gap-ordering sensitivity and exponential-envelope stability—are identified, with clear statistical tests and a numerically practical implementation.

The key methodological stance is that of *resource-bounded lawfulness*: only what can be computed reliably on real hardware, using well-understood numerical methods, counts as evidence for or against RH-stability in this model. This aligns naturally with a Meta-Relativistic view in which lawfulness emerges from recursively structured multiplicity under finite constraints.

Mathematical Appendix

A Operator Definitions and Basic Properties

A.1 Prime-chain Hilbert space

Let $p_1 < p_2 < \cdots < p_N$ denote the first N prime numbers. We consider the finite-dimensional Hilbert space

$$\mathcal{H}_N \cong \mathbb{C}^N,$$

with orthonormal basis vectors $\{|p_i\rangle\}_{i=1}^N$ satisfying

$$\langle p_i|p_j|p_i|p_j\rangle = \delta_{ij}, \qquad \sum_{i=1}^N |p_i\rangle \langle p_i| = I_{\mathcal{H}_N}.$$

A generic state is

$$|\psi\rangle = \sum_{i=1}^N c_i |p_i\rangle, \qquad \sum_{i=1}^N |c_i|^2 = 1.$$

A.2 Prime-diagonal operator

For a fixed real parameter $\alpha > 0$ we define

$$H_{\text{prime}} := \alpha \sum_{i=1}^N \frac{1}{p_i} |p_i\rangle \langle p_i|. \tag{15}$$

This is a diagonal operator in the $|p_i\rangle$ basis, with eigenvalues

$$\lambda_i^{(\text{prime})} = \frac{\alpha}{p_i}, \quad i = 1, \ldots, N.$$

Thus

$$H_{\text{prime}} |p_i\rangle = \frac{\alpha}{p_i} |p_i\rangle.$$

Since all eigenvalues are real, H_{prime} is self-adjoint. Its operator norm (spectral norm) is

$$\|H_{\text{prime}}\| = \max_{1\le i\le N} \left|\frac{\alpha}{p_i}\right| = \frac{\alpha}{p_1} = \frac{\alpha}{2}. \tag{16}$$

A.3 Gap-hopping operator

Let $g_i = p_{i+1} - p_i$ denote the prime gaps for $i = 1, \ldots, N-1$ and define normalized gaps $\tilde{g}_i$ by one of:

$$\tilde{g}_i^{(\max)} = \frac{g_i}{\max_j g_j}, \tag{17}$$

$$\tilde{g}_i^{(\log)} = \frac{\log(1+g_i)}{\log(1+g_{\max})}, \qquad g_{\max} = \max_j g_j. \tag{18}$$

In either case, $0 < \tilde{g}_i \leq 1$. We define the hopping operator

$$H_{\text{gap}} := \eta \sum_{i=1}^{N-1} \tilde{g}_i \left(|p_i\rangle \langle p_{i+1}| + |p_{i+1}\rangle \langle p_i| \right), \tag{19}$$

with $\eta \in \mathbb{R}$.

Self-adjointness. Each term

$$|p_i\rangle \langle p_{i+1}| + |p_{i+1}\rangle \langle p_i|$$

is self-adjoint, and the coefficients $\eta\tilde{g}_i$ are real, hence $H_{\text{gap}}^\dagger = H_{\text{gap}}$.

Matrix form. In the ordered basis $(|p_1\rangle, \ldots, |p_N\rangle)$, H_{gap} has matrix elements

$$\left(H_{\text{gap}}\right)_{ij} = \begin{cases} \eta\tilde{g}_i & \text{if } j = i+1, \\ \eta\tilde{g}_j & \text{if } i = j+1, \\ 0 & \text{otherwise.} \end{cases}$$

It is a real symmetric tridiagonal matrix with zeros on the diagonal.

Operator norm bounds. For each i, define the rank-2 operator

$$A_i := |p_i\rangle \langle p_{i+1}| + |p_{i+1}\rangle \langle p_i|.$$

Restricted to the span of $\{|p_i\rangle, |p_{i+1}\rangle\}$, A_i has matrix representation

$$A_i \leftrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which has eigenvalues ± 1. Hence $\|A_i\| = 1$. Using subadditivity,

$$\begin{aligned} \|H_{\text{gap}}\| &= \left\| \eta \sum_{i=1}^{N-1} \tilde{g}_i A_i \right\| \\ &\leq |\eta| \sum_{i=1}^{N-1} |\tilde{g}_i| \|A_i\| \leq |\eta| \sum_{i=1}^{N-1} \tilde{g}_i \leq |\eta|(N-1). \end{aligned} \tag{20}$$

A slightly sharper bound uses row sums. For any tridiagonal real symmetric matrix M,

$$\|M\| \leq \max_i \sum_j |M_{ij}|.$$

Here,

$$\sum_j |(H_{\text{gap}})_{ij}| \leq \begin{cases} |\eta|\tilde{g}_1 & i = 1, \\ |\eta|(\tilde{g}_{i-1} + \tilde{g}_i) & 2 \leq i \leq N-1, \\ |\eta|\tilde{g}_{N-1} & i = N, \end{cases}$$

so

$$\|H_{\text{gap}}\| \leq |\eta| \max\left\{ \tilde{g}_1, \max_{2 \leq i \leq N-1} (\tilde{g}_{i-1} + \tilde{g}_i), \tilde{g}_{N-1} \right\} \leq 2|\eta|. \tag{21}$$

The bound (21) is independent of N and typically much tighter than (20).

A.4 Time-dependent Hamiltonian and norm bounds

We define the full Hamiltonian by

$$H(t;\beta) := H_{\text{prime}} + \lambda(t;\beta) H_{\text{gap}}, \tag{22}$$

where

$$\lambda(t;\beta) = 1 + \sum_{n=1}^{K} \gamma_n e^{\beta t} \cos(t_n t + \phi_n), \qquad \gamma_n = \frac{\gamma_0}{\sqrt{n}}, \tag{23}$$

with $\gamma_0 > 0$, $t_n > 0$ the imaginary parts of the first K non-trivial zeros, and $\phi_n \in \mathbb{R}$ arbitrary phases.

A.4.1 Reality and self-adjointness

For each fixed t and $\beta \in \mathbb{R}$, $\lambda(t;\beta) \in \mathbb{R}$ since it is a finite sum of real exponentials times cosines. As H_{prime} and H_{gap} are self-adjoint, we have

$$H(t;\beta)^\dagger = H_{\text{prime}}^\dagger + \lambda(t;\beta) H_{\text{gap}}^\dagger = H_{\text{prime}} + \lambda(t;\beta) H_{\text{gap}} = H(t;\beta).$$

Thus $H(t;\beta)$ is self-adjoint for all (t,β).

A.4.2 Operator norm bound for $H(t;\beta)$

Using the triangle inequality and bounds above,

$$\begin{aligned} \|H(t;\beta)\| &\le \|H_{\text{prime}}\| + |\lambda(t;\beta)|\, \|H_{\text{gap}}\| \\ &\le \frac{\alpha}{2} + |\lambda(t;\beta)| \cdot 2|\eta|. \end{aligned} \tag{24}$$

We bound $\lambda(t;\beta)$ on a finite time interval $[0,T]$. For any $t \in [0,T]$,

$$\begin{aligned} |\lambda(t;\beta)| &\le 1 + \sum_{n=1}^{K} |\gamma_n| e^{|\beta| t} |\cos(t_n t + \phi_n)| \\ &\le 1 + e^{|\beta| T} \sum_{n=1}^{K} \frac{\gamma_0}{\sqrt{n}}. \end{aligned} \tag{25}$$

Let $C_K := \sum_{n=1}^{K} n^{-1/2}$, which satisfies $C_K \le 2\sqrt{K}$. Thus

$$|\lambda(t;\beta)| \le 1 + \gamma_0 e^{|\beta| T} C_K \le 1 + 2\gamma_0 e^{|\beta| T} \sqrt{K}. \tag{26}$$

Combining with (24) and (21) gives

$$\|H(t;\beta)\| \le \frac{\alpha}{2} + 2|\eta| \left(1 + 2\gamma_0 e^{|\beta| T} \sqrt{K}\right), \qquad t \in [0,T]. \tag{27}$$

In particular, for fixed $(\alpha, \eta, \gamma_0, K, T)$, the norm grows at most exponentially in $|\beta| T$.

B Existence and Unitarity of the Time-Ordered Evolution

B.1 General theorem for bounded self-adjoint $H(t)$

Consider a family of bounded self-adjoint operators $H(t)$ on a finite-dimensional Hilbert space $\mathcal{H}_N$ such that:

- The map $t \mapsto H(t)$ is continuous on a compact interval $[0,T]$.

- Each $H(t)$ is bounded, with $\|H(t)\| \leq M$ for some $M > 0$ and all $t \in [0, T]$.

Then the initial value problem

$$i\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle, \qquad |\psi(0)\rangle = |\psi_0\rangle \tag{28}$$

has a unique global solution $|\psi(t)\rangle$ for all $t \in [0, T]$, and the evolution is unitary, i.e. the solution can be written as

$$|\psi(t)\rangle = U(t, 0)|\psi_0\rangle, \tag{29}$$

where $U(t, s)$ is a two-parameter family of unitary operators satisfying

$$i\frac{d}{dt}U(t, s) = H(t)U(t, s), \qquad U(s, s) = I, \tag{30}$$

$$U(t, s)U(s, r) = U(t, r). \tag{31}$$

Moreover, $U(t, s)$ can be represented as a time-ordered exponential

$$U(t, s) = \mathcal{T}\exp\left(-i\int_s^t H(\tau)\,d\tau\right). \tag{32}$$

B.2 Application to the Zeta-Multiplicity Hamiltonian

In our case, for fixed parameters $(\alpha, \eta, \gamma_0, K, \beta)$ and a finite time horizon $[0, T]$, we have shown:

- $H(t; \beta)$ is self-adjoint for all t.
- $t \mapsto H(t; \beta)$ is continuous (indeed smooth) because $\lambda(t; \beta)$ is a finite linear combination of smooth functions.
- $\|H(t; \beta)\|$ is bounded uniformly on $[0, T]$ by the estimate above.

Thus, by the general theorem, the initial value problem

$$i\frac{d}{dt}|\psi(t)\rangle = H(t; \beta)|\psi(t)\rangle, \qquad |\psi(0)\rangle = |\psi_0\rangle \tag{33}$$

has a unique solution given by a unitary propagator $U_\beta(t, 0)$:

$$|\psi(t)\rangle = U_\beta(t, 0)|\psi_0\rangle,$$

and $\|\psi(t)\| = 1$ for all $t \in [0, T]$.

C Properties of Circular Phase Variance

C.1 Basic invariances

Recall the definition:

$$\theta_i(t) = \arg c_i(t), \tag{34}$$

$$R(t) = \frac{1}{N}\sum_{i=1}^{N} e^{i\theta_i(t)}, \tag{35}$$

$$\theta_{\text{mean}}(t) = \arg R(t), \tag{36}$$

$$\delta_i(t) = \arg\left(e^{i(\theta_i(t) - \theta_{\text{mean}}(t))}\right), \tag{37}$$

$$V(t) = \frac{1}{N}\sum_{i=1}^{N} \delta_i(t)^2. \tag{38}$$

[Global phase invariance] Let $|\psi'(t)\rangle = e^{i\varphi(t)}|\psi(t)\rangle$ for some real-valued function $\varphi(t)$. Then $\theta_i'(t) = \theta_i(t) + \varphi(t)$, but the circular phase variance satisfies $V'(t) = V(t)$ for all t where $R(t) \neq 0$.

Proof. Under a global phase shift,

$$\theta_i'(t) = \theta_i(t) + \varphi(t).$$

Then

$$R'(t) = \frac{1}{N}\sum_i e^{i\theta_i'(t)} = \frac{1}{N}\sum_i e^{i(\theta_i(t)+\varphi(t))} = e^{i\varphi(t)}R(t).$$

Thus $\theta'_{\text{mean}}(t) = \arg R'(t) = \arg R(t) + \varphi(t) = \theta_{\text{mean}}(t) + \varphi(t) \pmod{2\pi}$. Deviations transform as

$$\begin{aligned}\delta_i'(t) &= \arg\left(e^{i(\theta_i'(t)-\theta'_{\text{mean}}(t))}\right)\\ &= \arg\left(e^{i[(\theta_i(t)+\varphi(t))-(\theta_{\text{mean}}(t)+\varphi(t))]}\right)\\ &= \arg\left(e^{i(\theta_i(t)-\theta_{\text{mean}}(t))}\right) = \delta_i(t).\end{aligned}$$

Hence $V'(t) = \frac{1}{N}\sum_i \delta_i'(t)^2 = \frac{1}{N}\sum_i \delta_i(t)^2 = V(t)$. □

C.2 Boundedness

[Bounded variance] For any state $|\psi(t)\rangle$ with $R(t) \neq 0$,

$$0 \leq V(t) \leq \pi^2.$$

Proof. By construction, each deviation $\delta_i(t) \in (-\pi, \pi]$. Hence $\delta_i(t)^2 \leq \pi^2$ and $\delta_i(t)^2 \geq 0$. Taking the average,

$$0 \leq V(t) = \frac{1}{N}\sum_i \delta_i(t)^2 \leq \frac{1}{N}\sum_i \pi^2 = \pi^2.$$

□

D Remarks on Asymptotic Behavior and RH-Stability

D.1 Heuristic connection to β

The envelope

$$\lambda(t;\beta) = 1 + \sum_{n=1}^{K} \gamma_n e^{\beta t}\cos(t_n t + \phi_n)$$

has asymptotic behavior controlled by β on any sufficiently long finite window. For $\beta = 0$, the modulation is bounded,

$$|\lambda(t;0)| \leq 1 + \gamma_0 \sum_{n=1}^{K} \frac{1}{\sqrt{n}},$$

whereas for $\beta \neq 0$ one expects exponential growth or decay in magnitude (bounded only by the finite time horizon T in numerical experiments). Through the bound on $\|H(t;\beta)\|$, this envelope translates into bounds on the generator of dynamics.

While a rigorous proof connecting $\beta = 0$ to bounded $V(t)$ in the limit $T \to \infty$ would require significantly deeper analysis (and likely number-theoretic input), the finite-time operator norm bounds and the controlled exponential envelope suggest that any exponential growth in $V(t)$ for $\beta \neq 0$ is, at minimum, consistent with the stronger drive induced by $e^{\beta t}$.

D.2 Finite-dimensional truncation and regularization

All proofs above are carried out in a finite-dimensional setting. Questions about convergence as $N \to \infty$ or $K \to \infty$ would require additional regularization (e.g. zeta-regularized sums in the exponent, or truncation schemes tuned to the explicit formula). The current exposition explicitly treats N, K as finite and fixed, which is sufficient for the numerical protocol and for the defensive publication goals of this work.

E Summary of Key Bounds

For convenience, we collect key inequalities:

- $\|H_{\text{prime}}\| = \alpha/2$.
- $\|H_{\text{gap}}\| \leq 2|\eta|$.
- For $t \in [0, T]$,
$$|\lambda(t;\beta)| \leq 1 + 2\gamma_0 e^{|\beta|T}\sqrt{K}.$$
- Hence, for $t \in [0, T]$,
$$\|H(t;\beta)\| \leq \frac{\alpha}{2} + 2|\eta|\left(1 + 2\gamma_0 e^{|\beta|T}\sqrt{K}\right).$$
- Circular variance satisfies $0 \leq V(t) \leq \pi^2$ and is invariant under global phase shifts.

These bounds ensure well-posedness and unitarity of the finite-time evolution and provide an explicit quantitative handle on how the envelope parameter β and the truncation parameters (N, K, T) enter into the dynamical generator's norm.

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